

- MAYK DE ANDRADE, HUGO MARIANO, STEVE AWODEY, *An application of algebraic geometry tools to intuitionistic logic*.
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For any theory $\mathbb{T}$ in Classical FOL, we can construct its classifying Boolean pretopos $\mathcal{T}$. Models of this theory in any other Boolean pretopos $\mathcal{R}$ will be then given by Boolean pretopos functors $\mathcal{T} \rightarrow \mathcal{R}$. Makkai [1] has established an adjunction

$$\mathbf{BP}^{*\text{op}} \begin{array}{c} \xrightarrow{G} \\ \xleftarrow{F} \end{array} \mathbf{UG}$$

between the 2-category $\mathbf{BP}^*$ of Boolean pretopoi and natural isomorphisms between them and the 2-category $\mathbf{UG}$ of ultragroupoids. One side of this adjunction is essentially Gödel's Completeness Theorem.

This adjunction is part of a (lax) descent theorem, as it was shown by Zawadowski [?]. This fact was first glimpsed by Pitts [3] when he proved in a sequence of papers that Heyting pretoposes satisfy the interpolation property.

We aim to prove a variant of Stone duality and the descent theorem, but for Heyting pretoposes. One step towards this was developed in Morin's dissertation [2], providing an explicit embedding $e : \mathcal{H} \rightarrow \text{Hom}(\text{Coh}(\mathcal{H}, \mathbf{Set}))$ for any Heyting category. It remains necessary to check whether this embedding respects the ultracategory structure.

At the same time, we are looking for a similar embedding result for dependent type theory, a higher order well established version of intuitionistic logic.

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- [4] ZAWADOWSKI, M., *Descent and duality*, *Annals of Pure and Applied Logic*, vol. 71 (1995), no. 2, pp. 131–188.